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Theorem 0.1.

$$(\{y|\psi\} \text{ nonempty}) \longleftrightarrow (\exists y : \psi)$$

Proof.
($\Rightarrow$)

$$(\{y|\psi\} \text{ nonempty}) \implies (\exists y : \psi)$$

($\Leftarrow$)
Done.

Theorem 0.2. Any class is a subclass of class of all sets.

$$(C \subseteq V)$$

Proof.
Claim: $(\forall x \in C : x \in V)$.
Fix x. Assume $(x \in C)$. To show $(x \in V)$.

$$(x \in C) \implies (x \text{ set}) \quad \text{(by thm 0.10)}$$

$$(x \text{ set}) \ \& \ \top \implies (x \in V) \quad \text{(by def ??)}$$

Claim is proven.

$$(\forall x \in C : x \in V) \implies (C \subseteq V) \quad \text{(by def ??)}$$

Done.

Theorem 0.3.

$$(X \text{ set}) \longleftrightarrow (X \in V)$$

Proof.
($\Leftarrow$) Assume $(X \in V)$.

$$(X \in V) \implies (\exists W : X \in W).$$

$$(\exists W : X \in W) \implies (X \text{ set}).$$

($\Rightarrow$) Assume $(X \text{ set})$.

$$(X \text{ set}) \ \& \ \top[X/W] \implies (X \in \{x|\top[x/W]\})$$

$$(X \in \{x|\top[x/W]\} = \{x|\top\} = V) \implies x \in V$$

Done.

Theorem 0.4.

$$(A \subseteq B) \wedge (B \subseteq A) \longrightarrow (A = B)$$

Proof.
Claim: $(\forall x : x \in A \longleftrightarrow x \in B)$.
Fix x.
$(\Rightarrow)$
Assume $(x \in A)$. To show: $(x \in B)$

$$(x \in A) \ \& \ (A \subseteq B) \implies (x \in B) \quad \text{(by thm ??)}$$

$(\Leftarrow)$
Assume $(x \in B)$. To show: $(x \in A)$

$$(x \in B) \ \& \ (B \subseteq A) \implies (x \in A) \quad \text{(by thm ??)}$$

Claim is proven.

$$(\forall x : x \in A \longleftrightarrow x \in B) \implies (A = B) \quad \text{(by axm ??)}$$

Done.

Theorem 0.5.

$$(\emptyset \subseteq A)$$

Proof.
Claim:

$$(\forall x \in \emptyset : x \in A)$$

Assume $(x \in \emptyset)$.

$$(x \in \emptyset) \implies \bot$$
$$\bot \implies (x \in A)$$

Claim is proven.

$$(\forall x \in \emptyset : x \in A) \implies (\emptyset \subseteq A) \quad \text{(by def ??)}$$

Done.

Theorem 0.6.

$$\neg \exists w : w \in A \longrightarrow (A = \emptyset)$$

Proof.
Assume $\neg \exists w : w \in A$. To show: $(A = \emptyset)$.
$(\supseteq)$

$$(A \supseteq \emptyset) \quad \text{(by thm 0.5)}$$

$(\subseteq)$
Claim: $(\forall x \in A : x \in \emptyset)$
Fix x. Assume $(x \in A)$. To show: $\bot$

$$(x \in A) \implies (\exists x : x \in A)$$

$$(\exists x : x \in A) \ \& \ (\neg \exists x : x \in A) \implies \bot$$
$$\bot \implies (x \in \emptyset)$$

The claim is proven.

$$(\forall x \in A : x \in \emptyset) \implies (A \subseteq \emptyset) \quad \text{(by def of subclass ??)}$$
$$(A \subseteq \emptyset) \ \& \ (A \supseteq \emptyset) \implies (A = \emptyset) \quad \text{(by axm ??)}$$

Done.

Theorem 0.7.
$$\neg \exists w : w \in A \iff (A \text{ empty})$$

Theorem 0.8.
$$\exists w : w \in A \iff (A \text{ nonempty})$$

Theorem 0.9. Any element of a subclass is in the class.
$$(x \in A) \wedge (A \subseteq B) \longrightarrow (x \in B)$$

Proof.

$$(A \subseteq B) \implies (\forall x \in A : x \in B) \quad \text{(by def of subclass ??)}$$
$(x \in A) \ \& \ (\forall x \in A : x \in B) \implies (x \in B)$ (by restricted specification elimination)
Done.

Theorem 0.10.
$$(B \in A) \to (B \text{ set})$$

Proof.
Assume $(B \in A)$. To show: $(B$ set$)$.
$$(B \in A) \implies (\exists A : B \in A) \quad \text{(by right-exists-intro).}$$
$$(\exists A : B \in A) \implies (B \text{ set}) \quad \text{(introduced by set's definition).}$$

Done.

Theorem 0.11.
$$(w \in A) \wedge (!a : a \in A) \longrightarrow \{w\} = A$$

Proof.
$(\subseteq)$ Assume $(q \in \{w\})$.
$$(q \in \{w\}) \implies (q = w).$$
$$(w \in A) \ \& \ (q = w) \implies (q \in A).$$
$(\supseteq)$ Assume $(a \in A)$.
$$(!a : a \in A) \ \& \ (a \in A) \ \& \ (w \in A) \implies (a = w).$$
$$(a = w) \implies (\{a\} = \{w\})$$
$$(a \in \{a\}) \ \& \ (\{a\} = \{w\}) \implies (a \in \{w\})$$

Done.